

Use of Drone Data for Spatial Analysis of Tree Clustering and Water Stress



Master thesis proposal



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1. Context and Background

Climate change models forecast a rise in the frequency, intensity, and duration of droughts in the future. As a result of anthropogenic climate change, forest ecosystems are encountering substantial challenges like altered precipitation patterns, droughts and a loss of biodiversity (McDowell et al., 2020). Drought is a significant environmental stressor that can have profound impacts on forest ecosystems by affecting tree growth, mortality rates and overall ecosystem functioning (Puchi et al., 2021). Therefore, it is important to get better insights of how forests cope with increasingly severe conditions (Brodribb et al., 2020). Severe water stress conditions in which the available water supply is limited or insufficient for the needs of trees have been observed to instigate mechanisms such as hydraulic failure, carbon starvation, and heightened susceptibility to biotic factors, all increasing tree mortality (McDowell et al., 2008) (Laita et al., 2023). This is concerning since forest biomass is having an important role in the fight against climate change by absorbing carbon dioxide out of the air (Avitabile et al., 2024).

In conditions of water scarcity by plants, a reduction of transpiration in turn directly affects leaf temperature (Jackson et al., 1981) (Sulman et al., 2016). When water is available transpiration can lead to canopy temperatures far below air temperatures (Snakin et al., 2001). Because of this canopy temperature can be used to find changes in stomatal regulation and to get insights on water stress (Šmigaj et al., 2023). But many aspects have influence on water stress. Like, how much shadow a tree has, the height of trees and the species. Shadow from taller trees, for example, can lead to a lower mortality ratio since shadow leads to a lower leaf temperatures and thus less transpiration (Ma et al., 2023). Next to this diurnal changes in water exchange can give valuable insights into the water status of trees (Baldocchi & Wilson, 2001). If water uptake or transpiration rates are different from expected patterns, it could indicate stress. Stressed trees have contrasting diurnal patterns compared to unstressed trees, with differences in stem water content and transpiration rates (Liu et al., 2023). Although an important note is that these diurnal variations are expected to change because of climate change (Z. Zhang et al., 2023). It should be noted here that Liu et al. (2023) and Zhang et al. (2023) focusses on totally different ecosystems. Understanding individual tree's response to drought stress and how trees affect each other is crucial for assessing the impact of climate change, since trees play a significant role in Earth's ecosystems and climate regulation. Forest management can manipulate tree species diversity or the amount of trees in a forest. This affects stand structure and leads to changes in forest services and functions (Dieler et al., 2017). Such knowledge can be used to have influence on the water situation of the forest and improve resilience.

Earth observation has been used for forest monitoring and land cover mapping for over 50 years (Deshayes et al., 2006). Since the start of this a lot of new technologies have been developing. At the beginning the common method to collect data was satellites, in recent years unmanned aerial vehicles (UAV's) are becoming more widely used (Latifi & Valbuena, 2019). Remote sensing has emerged as a valuable asset in acquiring data on forest health, offering the possibility to collect data on large scale in a non-destructive way. However, when utilizing carrier platforms like satellites or crewed airborne systems, the resolution in spatial and/or temporal accuracy often falls short. Conversely, the spatio-/ temporal resolution may be too coarse to offer detailed insights into the subtle physiological and morphological responses of trees to stress. These responses can manifest at an intermediate level, affecting specific individuals or groups of trees. UAV's offer a solution to this problem. With their capacity to capture data at high spatial and temporal resolutions in inaccessible terrain, UAV's present a fresh avenue for gaining deeper insights into forest ecosystems (Ecke et al., 2024).

2. Research needs

Droughts lead to higher tree mortality which alters the structures and spatial arrangement of forests. Although It has been posited that larger trees exhibit a larger sensitivity to drought conditions than smaller trees (Bennett et al., 2015), it can be seen as controversial since some studies have different results. For example, (Ma et al., 2023) showed that larger trees acquire water at the expense of smaller trees during droughts. What can be stated is that individual tree size (e.g., crown area, height, diameter) influences mortality of trees by regulating water demands and supply (Ma et al., 2023). Topography together with broader scale climate factors, create spatially heterogeneous fine-scale environments (Fan et al., 2019), which play a crucial role in shaping individual tree responses to drought. The systematic fluctuations in water availability is a significant mechanism shaping the spatial patterns of vegetation biomass (Thomas et al., 2024).

While remote sensing techniques have been extensively employed to study drought at the forest level, limited attention has been given to utilizing remote sensing techniques to investigate trees at individual tree level. This is surprising since incorporating modern UAV and sensor technology can gradually improve, or even replace costly and labour intensive ground based assessments (Ecke et al., 2024), and shed new light on forest functioning. LiDAR technology offers a powerful tool for monitoring canopy structure, multispectral imagery offers the possibility to calculate different vegetation indices, while thermal data can be effectively utilized to measure canopy temperature (Ma et al., 2023). Previous UAV studies mainly focused on drought stress of individual trees, without looking at the interplay between them. For example thermal imaging for drought stress has been explored by Krause and Sanders (2023) who used UAV-based thermal imaging to map forest stands and individual trees affected by drought stress. They showed that variations in tree water deficit could successfully be modelled, even in the absence of actual drought events. Adeline et al. (2024) used LiDAR 3D point clouds collected with a UAV for mapping and monitoring of Mediterranean oaks focussing on leaf and canopy scale. They showed how different types of data could be utilized for various ecological applications. Although the possibilities and effectiveness of data collection with UAV's to asses droughts on tree and forest level is shown by these studies, there remains a notable gap in the scientific literature regarding the interplay between trees with each other in the context of water stress. Gaining new insights into this aspect is important for both present and future forest management strategies.

3. Research objectives and Research questions



3.1 Research objective

This research aims to investigate how the forest canopy structure influences tree water stress levels. First a literature review will be conducted to identify how remote sensing stress indices detect water stress in trees. Additionally, the literature review will explore how LiDAR data can be used to characterize forest canopy structure (RQ1 and RQ2). This information will be used to analyse the influence of spatial tree arrangement on different remote sensing stress indices and to examine the relationship between diurnal patterns of stress indices and forest canopy (RQ3 and RQ4).

3.2 Research questions

1. How can water stress on trees be detected by remote sensing stress indices?
2. How can LiDAR data be used to characterize forest canopy structure?
3. How does spatial tree arrangement influence different remote sensing stress indices?
4. What is the relationship between diurnal patterns of stress indices and forest canopy structure?

4. Methods

In this section we will dive into the methodology of this research project. An overview of the steps that will be conducted in this thesis is illustrated in figure 1. The yellow blocks which are data blocks, and the blue blocks which are processing steps. Additionally, the steps that lead to answering the research questions are grouped and circles show intermediate products. This thesis is divided into two stages: a literature review segment where research questions 1 and 2 are addressed, followed by an analytical part. In the analytical phase, insights derived from the literature review pertaining to research questions 1 and 2 are integrated with UAV data to address research questions 3 and 4.

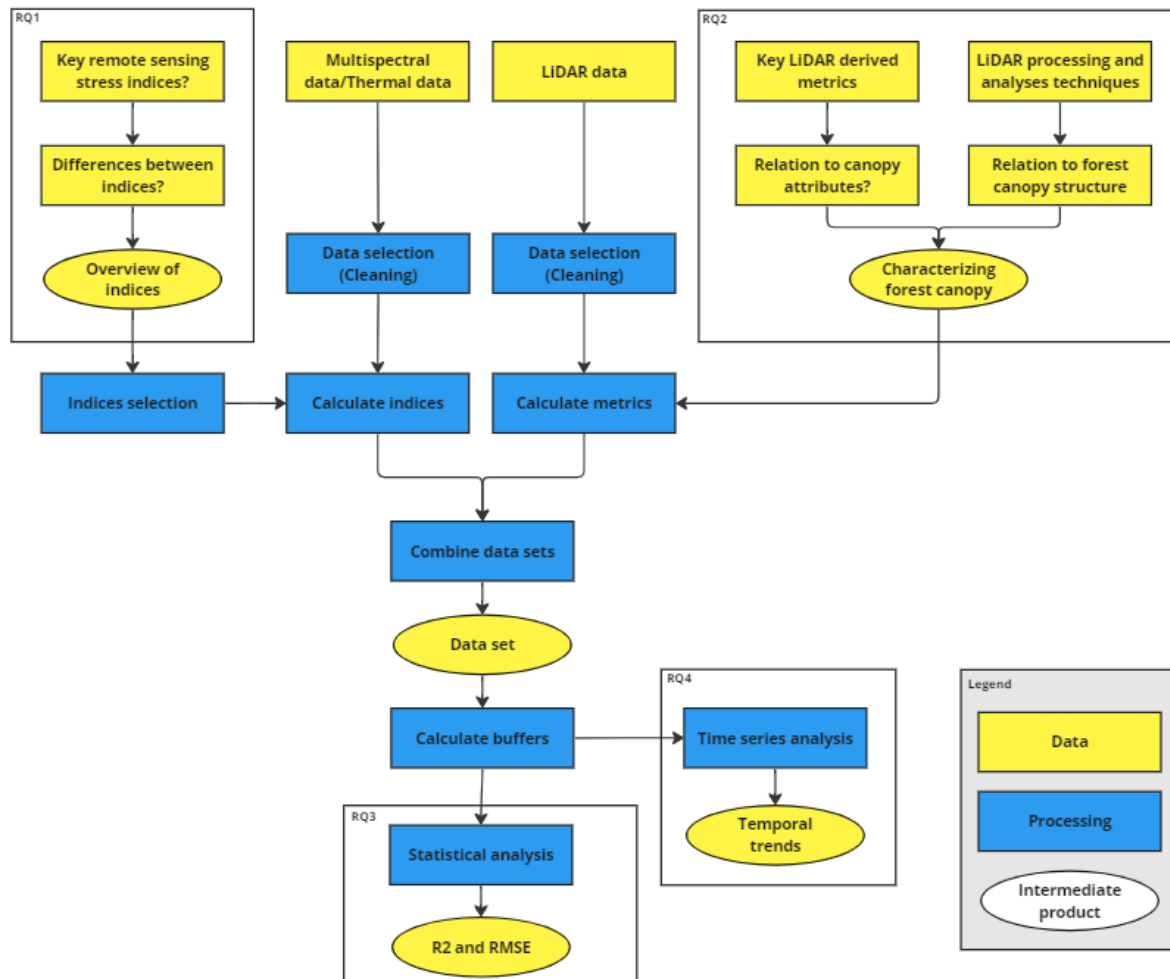


Figure 1: Overview of the different steps and their relation to the different research questions

4.1 Data

There are three plots available which regularly get scanned with a UAV. They are located on two different locations in The Netherlands and consist of different kind of trees. All sites are located on flat terrain at 30~70 meter above sea level. They are characterized by poor sandy podzols, such poor conditions dominate approximately 80% percent of The Netherlands and 45% of the forests across central Europe.

- Scots Pine (Zwolve bos)
- Beech (Uchelen)
- Douglas fir (Zwolve bos)

In table 1 the stand age (GY), diameter (ADD), tree height (ADH) of the different plots is shown. It can be stated that the plots themselves are quit uniform, except of the spatial arrangement and density.

Table 1: Information different plots

Set	MS	GY	ADD	ADH	Description
1	SP	1964	33,3	18,8	Scots pine Zwolve Bos (1SP)
2	BE	1918	69,9	25,6	Beech Ugchelen (2BE)
1	DG	1942	65,7	41	Douglas fir Zwolve Bos (1DG)

The plots are approximately 1-1.2 Ha in size and are divided into four sections. One clear cut (where all trees removed), a shelterwood system (with 80% removed), a high thinning zone (with 20% removed) and a control plot in which no trees are removed. A single day per plot will be used. Approximately the plots got scanned at 5 different times on a single day.

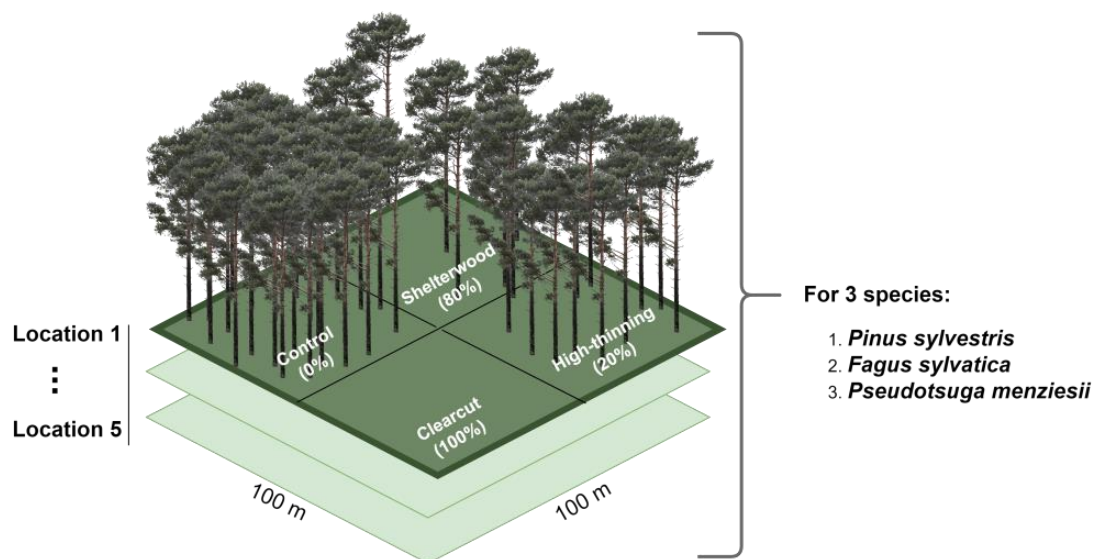


Figure 2: Side view of one of the plots (Meijers, 2023)

The dataset consists of multispectral, thermal and LiDAR data. The specific data to be utilized depends on the findings of the literature review. Since the selection of remote sensing indices and canopy metrics have to be made based on the knowledge gained with research question 1 and 2. It can be stated that temporal data will be used of different UAV flights on a single for research question 4.

4.2 Literature review

To answer research question 1 and 2 a literature review will be done. To select papers the snowballing procedure will be used, which is illustrated in figure 3 (Wohlin, 2014). To apply snowball sampling a small set of papers, ideally review articles, relevant to the research question will be collected to start with. The search will then be expanded by backward snowballing in which there will be looked at relevant papers in the literature list, and forward snowballing in which there will be looked at which papers cited the paper being examined. In this research it will not be an exhaustive search in terms of publications. There will be focussed on the methods/metrics and the search will be ended when saturation is reached.

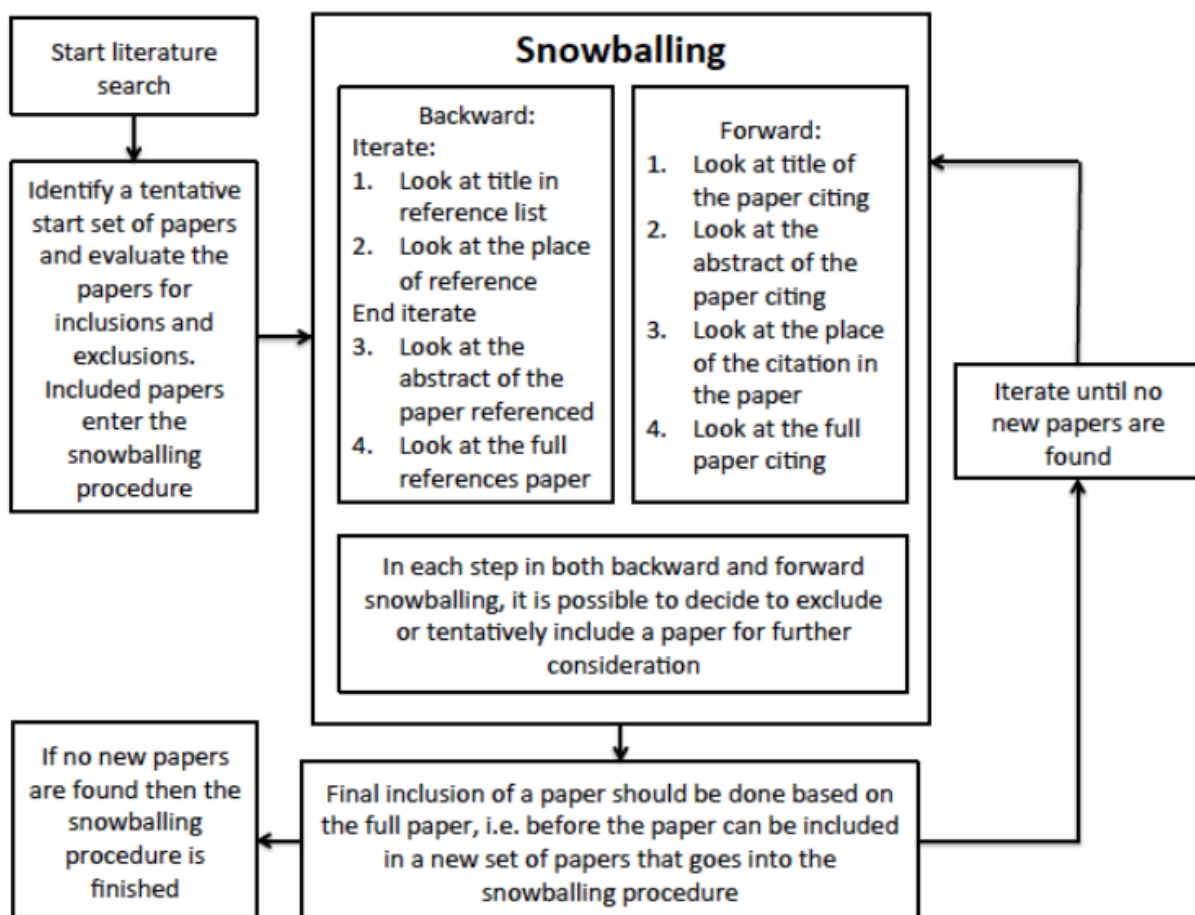


Figure 3: Snowballing procedure (Wohlin, 2014)

Research question 1:

To answer research question 1, first an overview of the key indices to detect water stress will be created in which a divergence between multispectral and thermal indices can be found. The strengths, weaknesses and differences between indices will also be identified. This will help establish how water stress on trees can be detected with remote sensing indices.

Research question 2:

The next step will be to answer research question 2. Forest canopy structure is a broad term with many different aspects, including vertical distribution of layers within canopy, spatial distribution of trees, forest cover and tree-level structural parameters. In this thesis, due to time limitations, the focus will be put on the spatial arrangement of trees and the forest cover. To answer the second research question, it is essential to examine what kind of metrics can be derived from LiDAR to capture the forest canopy structure, and the methods used to analyse it. Subsequently, these LiDAR metrics must be linked to aspects of the forest canopy and attributes. By doing this it can be answered how LiDAR can be used to characterize forest canopy structure.

4.3 Data Analyses

Research question 3:

In research question 3 an analysis will be done, in which the results of the literature review will be used. One or multiple stress indices chosen after answering research question 1, will be calculated for the forest plots. After which metrics on the trees will be calculated using LiDAR data. Next to this metrics of tree spacing will be calculated, by using nearest neighbour distance, spatial autocorrelation, Ripley's K function. Buffers of different sizes will be calculated on trees. This to find out how many trees are in the area of other trees, and how close they are. A pattern might be characterized between stress indices and trees. After this a regression analysis will be done to identify relations between spatial forest metrics and the water stress indices. The results will be an R^2 which measures the strength of correlation. These can be compared with each other to answer how tree arrangement influences remote sensing stress indices.

Research question 4:

Research question 4 focusses on the relationship between diurnal patterns of stress indices and forest canopy metrics. There will be looked if diurnal patterns of stress indices exist. A time series over a day will be analysed to find spatial temporal trends on the plots. There will be looked at fluctuations over the course of the day and how it is related to different forest canopy conditions.

Table 2: Gantt chart, thesis planning

[illegible]

6. Feasibility

6.1 Time

Time is a limiting factor in the thesis. Estimating the duration required for the data processing is difficult due to my limited experience with LiDAR data processing. If time constraints arise, it may be needed to make some adjustments. There are several possibilities during this thesis research for time control:

- After research question 1, the selection of indices may involve prioritizing on one indices instead of multiple indices. Even if alternative indices could potentially enhance the study.
- Data from three distinct plots is available, during the data processing phase, a decision may be made to exclude plots to save time.
- Leave research question 4 out and focus on the relationship of stress indices and different species using all the plots.

6.2 Data

The data is to be made available by Arjen de Jonge. Given my limited experience with LiDAR data and integrating different kind of data, a lot of time is planned for the data processing part. Furthermore, the feasibility of processing all the data on my personal computer is uncertain. Which means I might need to use a computer of the Wageningen University, which will not be a problem but should be taken into account.

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